Lecture plan · for the lecturer

Asymptotic analysis

Programming · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Use O, Θ and Ω precisely.
- Solve recurrences.
- Teach growth by measuring.

Activity: Time it, then prove it

1. Warm-up (5 min): Ask students to predict how long a double loop takes on ten million records.
2. Measure (20 min): Pairs time insertion sort and merge sort on growing inputs and plot both.
3. Define (15 min): Introduce O, Θ and Ω to explain the curves.
4. Prove (15 min): Draw merge sort's recursion tree and apply the master theorem.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Where did the curves cross?
- Why do constants vanish?
- Which misconception did you hear?

Exit ticket

- Write one true and one false statement using Big-O.

Try it in your next lecture
Give pairs insertion sort and merge sort and inputs from 10 to 10,000 items (merge sort alone up to a million); they time both, plot the results, find where the curves cross and decide which curve matches n log n.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **A student says this linear search is $O(n^2)$. Is that true?**

```
def find(xs, target):
    for i in range(len(xs)):
        if xs[i] == target:
            return i
    return -1
```

◯ False ◯ True, but not tight ◯ Only for sorted lists

2 **Find the constants.**

To show $3n^2 + 10n + 7 = O(n^2)$ with $c = 4$, the smallest whole n_0 that works is _____ , and merge sort runs in $\Theta(n$ _____ $n)$.

Word bank: exp · 11 · log · 7

3 **Which statements are true? Choose all that apply.**

☐ $2^n = O(n^{100})$ ☐ Two nested loops over n are $O(n^2)$ ☐ Binary search is $\Theta(\log n)$ in the worst case

☐ $n \log n = O(n^2)$

4 **On small inputs, an $O(n^2)$ algorithm can be faster than an $O(n \log n)$ one.**

◯ True ◯ False

5 **Match each recurrence to its solution.**

$T(n) = 2T(n/2) + n$	• • • •	$\Theta(\log n)$
$T(n) = T(n/2) + 1$	• • • •	$\Theta(n)$
$T(n) = 4T(n/2) + n$	• • • •	$\Theta(n \log n)$
$T(n) = T(n - 1) + 1$	• • • •	$\Theta(n^2)$

Answers · for the lecturer

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1 **True, but not tight**

O is an upper bound. In the worst case Θ(n) is the tight one; in the best case, when the item comes first, it is Θ(1).

2 **To show $3n^2 + 10n + 7 = O(n^2)$ with c = 4, the smallest whole n_0 that works is 11, and merge sort runs in Θ(n log n).**

$4n^2 \geq 3n^2 + 10n + 7$ needs $n^2 \geq 10n + 7$, first true at n = 11 ($121 \geq 117$).

3 **n log n = $O(n^2)$; Binary search is Θ(log n) in the worst case; Two nested loops over n are $O(n^2)$**

Every exponential eventually outgrows every polynomial, so 2^n is not $O(n^{100})$.

4 **True**

Constants can make the n^2 algorithm faster on small inputs; insertion sort beats merge sort on short arrays.

5 **$T(n) = 2T(n/2) + n$ — Θ(n log n)**
$T(n) = T(n/2) + 1$ — Θ(log n)
$T(n) = 4T(n/2) + n$ — $\Theta(n^2)$
$T(n) = T(n - 1) + 1$ — Θ(n)

Compare f(n) with n^(log_b a), or draw the recursion tree.

✎ **For you to think about**

Where do your students first meet Big-O, and do they measure anything before seeing the definition?